Bayesian Optimization

Chapter Summary – Sections 5–8

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Overview: Sections 5–8

1. **Acquisition Functions** – LCB, PI, EI: balancing exploration and exploitation
2. **Acquisition Function Optimization** – How to maximize the (cheap) ACQF
3. **Noisy BO** – Handling stochastic evaluations
4. **Parallel BO** – Proposing batches of points for parallel compute

Running theme: the surrogate gives us mean $\hat{c}(\boldsymbol{\lambda})$ and uncertainty $\hat{\sigma}(\boldsymbol{\lambda})$ – acquisition functions turn these into a single proposal rule.

5. Acquisition Functions

Goal: trade off **exploitation** (low $\hat{c}$) and **exploration** (high $\hat{\sigma}$). Optimizing $\hat{c}$ alone $\Rightarrow$ pure exploitation, local minima.

ACQF	Formula	Key property
LCB	$\hat{c}(\boldsymbol{\lambda}) - \tau\hat{\sigma}(\boldsymbol{\lambda})$	Simple; τ controls trade-off; regret bounds for GP
PI	$\Phi\left(\frac{c_{\min} - \hat{c}(\boldsymbol{\lambda})}{\hat{\sigma}(\boldsymbol{\lambda})}\right)$	Prob. of beating $c_{\min}$; ignores improvement size; greedy
EI	$\mathbb{E}[\max\{c_{\min} - Y(\boldsymbol{\lambda}), 0\}]$	**Default**: rewards magnitude of improvement; closed form for Gaussian posterior

- All three equal 0 at design points for a noise-free (interpolating) GP
- EI closed form exists
- Under mild conditions, BO + GP + EI **converges to the global optimum** (Bull 2011)

6. Acquisition Function Optimization

Problem: $u(\lambda)$ is cheap but non-convex, multi-modal, and gets more rugged as more data is collected. This is still a hard inner optimization problem.

Setting	**Method**	**Notes**
Numeric $\mathbf{\Lambda}$	Multi-start L-BFGS-B	Analytic ∇u available for GP + standard kernels + ACQFs
Numeric $\mathbf{\Lambda}$	CMA-ES (IPOP/BIPOP restarts)	Gradient-free; adapts co-variance to local curvature; robust on ill-conditioned functions
Mixed / hierarchical $\mathbf{\Lambda}$	Iterated one-exchange local search (SMAC)	Change exactly one HP per step; handles categorical & conditional HPs

7. Noisy Bayesian Optimization

Setting: we observe $y = c(\boldsymbol{\lambda}) + \epsilon,\ \epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ – the true $c(\boldsymbol{\lambda})$ is latent.

Three adjustments needed:

1. **Surrogate** – use a *nugget GP* (add $\sigma_\epsilon^2 \mathbf{I}$ to kernel matrix): posterior *smooths* through the noisy data; $\hat{\sigma}^2(\boldsymbol{\lambda}^{(t)}) > 0$ even at design points. σ_ϵ^2 learned via marginal likelihood.
2. **Acquisition function** – $c_{\min} = \min_t y_t$ is noisy; lucky outliers dominate:
   - **AEI** (Augmented EI): plug-in $c^*_{\min}$ + penalty proportional to $\sqrt{\sigma_\epsilon^2}/\sqrt{\hat{\sigma}^2 + \sigma_\epsilon^2}$
   - **NEI** (Noisy EI): Monte Carlo over plausible noiseless values at design points
   - **LCB**: no patch needed – contains no $c_{\min}$ term
3. **Incumbent selection** – avoid noisy empirical best; prefer model-based $\arg\min_t \hat{c}(\boldsymbol{\lambda}^{(t)})$ or risk-averse $\arg\min_t \hat{c}(\boldsymbol{\lambda}^{(t)}) + \beta\, \hat{\sigma}(\boldsymbol{\lambda}^{(t)})$

8. Parallel BO

Goal: propose a batch $(\boldsymbol{\lambda}^{[1]}, \ldots, \boldsymbol{\lambda}^{[q]})$ to exploit q parallel workers. Naive top-q maximizers of u collapse to near-duplicates – explicit diversification is required.

Method	Key idea
qEI	Joint expected improvement over batch; MC + reparameterization trick
Kriging believer	Sequentially fantasize outcome at $\boldsymbol{\lambda}^{[k]}$ (posterior mean) before proposing $\boldsymbol{\lambda}^{[k+1]}$; fantasy lowers uncertainty there
q-LCB	Draw exploration weight $\tau_k \sim \text{Exp}(1)$ per slot; different weights spread proposals automatically; embarrassingly parallel
Thompson sampling	Sample q posterior paths; propose $\arg\min$ of each

Asynchronous BO (ADBO): workers react per-completion; pending (in-flight) configs are *fantasized* locally before each new proposal.